ABTSEGA TESFAYE CHUFARE

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Professional Summary

Security-focused software engineer and M.S. Cybersecurity candidate (May 2027) with hands-on work in cloud and application security, Python automation, AWS/Terraform infrastructure risk, ML-based threat detection, and AI/agent authorization. Developed and tested Python security tools spanning IaC scanning, agent tool-call policy enforcement, CloudTrail-style detection, and a multi-tenant FastAPI service. Experienced with security assessments, vulnerability remediation, CI/CD validation, API development, and clear technical documentation. Seeking early-career roles in Cloud Security, DevSecOps, Application Security, Python/Software Engineering, AI Security, and Detection Engineering.

Technical Skills

Cloud Security: AWS IAM, VPC, EC2, CloudTrail; Security Hub, AWS Config, GuardDuty lab exposure; RBAC, MFA, least privilege, logging, KMS concepts

Application Security: OWASP Top 10, threat modeling, secure SDLC, vulnerability assessment, penetration testing, SAST/DAST/SCA concepts, CVE/CWE/CVSS, remediation validation

Networks & Detection: TCP/IP, DNS, VPNs, firewalls, network segmentation, packet analysis, Wireshark, Snort, intrusion detection, network monitoring, incident triage

Governance & Risk: NIST CSF, NIST SP 800-53, CIS Controls, SOC 2, ISO 27001, FedRAMP Moderate control-mapping familiarity; cloud risk assessment

Software & APIs: Python, SQL, TypeScript/JavaScript, Java, C/C++; FastAPI, REST APIs, SQLite, backend services, Git/GitHub, unit and integration testing

DevSecOps & IaC: Terraform, Docker, Linux, CI/CD, GitHub Actions workflows, SARIF, configuration hardening, security gates, secrets-handling concepts

AI / ML Security: Tool-call policy enforcement, RBAC, tenant isolation, prompt-injection testing, LLM evaluation, scikit-learn, TensorFlow, DistilBERT

Other: SQLite, SQL parameterization, API authorization, threat modeling, JSON/JSONL analysis, logging and audit evidence, reproducible test fixtures

Work Experience

Mercor | Remote | **Contract Evaluation Specialist, AI Safety, Security & Risk** | Jul 2026 - Present

- Assess AI-generated code and cybersecurity outputs for correctness, secure behavior, exploitability, and policy compliance; reproduce edge cases and produce evidence-backed findings and developer-facing remediation guidance.
- Design adversarial test cases for LLM reliability, prompt/security failures, unsafe tool interactions, authorization mistakes, and agent workflows; use Python-assisted review to make evaluations repeatable.

The George Washington University | Washington, DC | **Cybersecurity Researcher** | Aug 2025 - Present

- Developed Python/ML pipelines to detect Terraform infrastructure misconfigurations and risky security patterns in 52,568 public projects; achieved PR-AUC 0.9002 with a security-aware fusion model.
- Studied AWS IAM, cloud data protection, identity, network exposure, logging, and secure configuration practices; translated technical results into experiments, control recommendations, publications, and presentations.

KIIT University, IoT Research Lab | Bhubaneswar, India | **Research Assistant** | Mar 2022 - Mar 2025

- Built and tested Python/C++ and cloud-connected IoT prototypes; troubleshot APIs, network behavior, software interfaces, configuration changes, and data-flow reliability with multidisciplinary collaborators.
- Documented system designs, experimental results, defects, and validation steps; supported testing and integration across software, cloud infrastructure, and data workflows.

KIIT University | Bhubaneswar, India | **AI & Cybersecurity Summer Intern** | Aug 2024 - Sep 2024

- Developed and evaluated Random Forest, SVM, and neural-network models on network telemetry for malicious-traffic classification, achieving 92% classification accuracy with reproducible Python experiments.

Prodigy InfoTech | Remote | **Cybersecurity Technology Intern** | May 2023 - Jul 2023

- Performed structured application/network security assessments; identified and helped remediate 15+ critical vulnerabilities, documented OWASP-aligned risk categories, and supplied retest and remediation guidance.

Selected Engineering Projects

ML-Based Infrastructure-as-Code Misconfiguration Research | Python, Terraform, DistilBERT, scikit-learn | 2026

- Developed preprocessing, chunking, model training, and evaluation workflows for security misconfiguration research spanning 52,568 Terraform projects; compared baseline and fusion approaches.
- Achieved research PR-AUC 0.9002 (fusion) versus 0.8862 (DistilBERT baseline); F1 0.8380; paper accepted at IC3T 2026. Results are from research, not the new scanner below.

IaC Guard: Terraform Misconfiguration Scanner | Python, AWS, Terraform, JSON, SARIF | 2026

- Implemented an explainable static scanner for literal Terraform AWS resources; flags public SSH ingress, S3 public ACLs, public RDS, IAM wildcard patterns, EBS encryption, and disabled public ACL blocking.
- Built file/line evidence, CWE-linked remediation output and JSON/SARIF exports for security review workflows; added synthetic hardened/vulnerable fixtures and automated tests.

Agent Policy Lab: AI Tool-Call Authorization | Python, RBAC, Tenant Isolation, Audit | 2026

- Built a deterministic policy gate for simulated AI-agent tools with allowlisted actions, role-based permissions, tenant-bound checks, input-schema limits, and deny-by-default decisions.
- Tested cross-tenant access, role escalation and prompt-injection cases; created a hash-chained local audit log for tamper-detection demonstrations, with explicit production limitations.

Cloud Detection Lab | Python, CloudTrail-style JSONL, MITRE ATT&CK | 2026

- Implemented timestamp-normalized event analysis and sliding-window correlation for AWS console login failures, unsafe security-group exposure, and administrator policy attachment.
- Produces analyst-ready alerts with severity, principal, source IP, ATT&CK mapping and triage guidance; validated with synthetic event fixtures and unit tests.

Secure Ticket API | FastAPI, SQLite, Python, REST, AppSec | 2026

- Created a tested multi-tenant API with role-gated ticket creation, tenant-scoped reads to prevent IDOR, parameterized SQL, schema validation, response models and basic security headers.

AWS Cloud Security Risk and Controls Assessment | AWS, IAM, VPC, NIST CSF | 2026

- Examined identity/access, network exposure, logging and sensitive-data risks in cloud lab scenarios; documented control gaps, mitigation priorities and cloud configuration guidance.
- Connected defensive configuration checks with practical remediation, least privilege, monitoring and security-review documentation.

Open-Source Software Contributions and Testing | TypeScript, GitHub, CI/CD | 2026

- Prepared tested open-source feature updates for calendar export and URL-state filtering, including input validation, stable IDs, tests, and documentation.
- Ran passing automated test suites (13 and 15 tests) and production builds; used Git workflows, debugging and developer-facing test evidence.

Education

The George Washington University | M.S. Cybersecurity in Computer Science | Expected May 2027 | GPA: 3.8/4.0

KIIT University | B.Tech Computer Science & Engineering | May 2025

Relevant Coursework / Labs: Network and Cloud Security, Secure Software Development, Application Security, Computer Networks, Operating Systems, Cloud Computing, Databases/SQL, Distributed Systems, Machine Learning, AI Security, Security Risk Management, AWS Academy Cloud Foundations.

Certifications and Research

Certifications / Training: CompTIA Security+ (SY0-701); Cisco CyberOps Professional; Cisco CCNA Networking; AWS Academy Cloud Foundations; Google Cybersecurity.

Conference Paper: "Learning Security Patterns in Infrastructure-as-Code for Misconfiguration Detection," IC3T 2026 (accepted).

IEEE S&P Workshop Poster: "Cyberbiosecurity Risk Assessment Framework for Genomic Data Storage in Cloud Environments," CyberBio2026.